

THE FIVE CLOCKS

京张五速

A Chassis for the Next Hundred Years

执行摘要

- ▶ The mismatch: AI turns over in 0.5–2 years; the chassis must last 100 years
- ▶ Method: solid slow layers + standard interfaces for fast layers
- ▶ Site law: the alignment outlives every train (1905–2026)
- ▶ Deliverable: 5 layers + 3 laws + inverse procedures + 25-y Assembly
- ▶ A child born in 2026 turns 25 at the 2051 Assembly



Method: Century Design ≠ Century Prediction

方法论：百年设计 ≠ 预测百年

康泽恩 Conzen 1960	Streets outlive plots outlive buildings
布兰德 Brand 1994	Six shearing layers at different paces
罗西 Rossi 1966	Permanences that withstand time
哈布瑞肯 Habraken 1961	Supports separated from infill (SI)
北京总规 Beijing masterplan	Strategic reserve: not deciding is a decision



Shared conclusion: build the un-guessable layers solidly; give the guessable layers standard interfaces. This proposal refuses to predict the technology of 2126 — the refusal is the method.

- Three laws: lease not occupy / fast never rewrites slow / inverse difficulty
- "The robots need it" is never a reason to modify the street network
- L5 default exit: inaction removes the device

The Five-Layer Matrix

五层矩阵

Layer/Clock	Period*	Instrument	Procedure
L1 河钟 River	500 年+ y	Blue-green protection line 蓝绿保护线	Not amendable 不可议
L2 线钟 Line	100 年 y	Alignment-protection plan 线位保护图则	25-y Repave Assembly 25 年复线大会
L3 块钟 Block	50 年 y	Grain guideline + easement 颗粒度导则+地役权	Public hearing 听证
L4 面钟 Section	15–30 年 y	Elastic sections + device band 弹性断面图集+设备带	Planning permit 规划许可
L5 器钟 Device	0.5–5 年 y	Lease + ledger + deposit pool 租约制+账牌+押金池	Filing · default exit 备案·默认退出

* Periods are design assumptions (A-CLOCK-001). The naming echoes Dazhongsi: the Yongle Bell tolls hours; these clocks toll eras.

Diagnosis: Grain and Sections

诊断：颗粒与断面

The double cut: the R-02 census has been executed (2026-08-21 OSM snapshot, ODbL). The belt is cut twice — lengthwise by at-grade rail, crosswise by compounds. Topology basis, not legal enclosure; provisional bbox.

- ▶ 344 blocks, bimodal: median 0.662 ha; 66 superblocks (19.2%) hold 84.8% of area
- ▶ At-grade Jingbao line: 8 crossings over 6,504.7 m (1/813 m); max gap 1,110.2 m
- ▶ 3 reserved corridors target the worst 3.24 km band (2 crossings only)
- ▶ The L4 lesion: aged sections without device band or conduit headroom
- ▶ Renewal: retain first, demolish the exception (Beijing regulation)

指标—证据链 Metrics—Evidence Chain



证据链 Evidence chain



R-02 街坊颗粒普查协议 Block-grain census protocol



三项视觉核验指标 Three visual-gate metrics



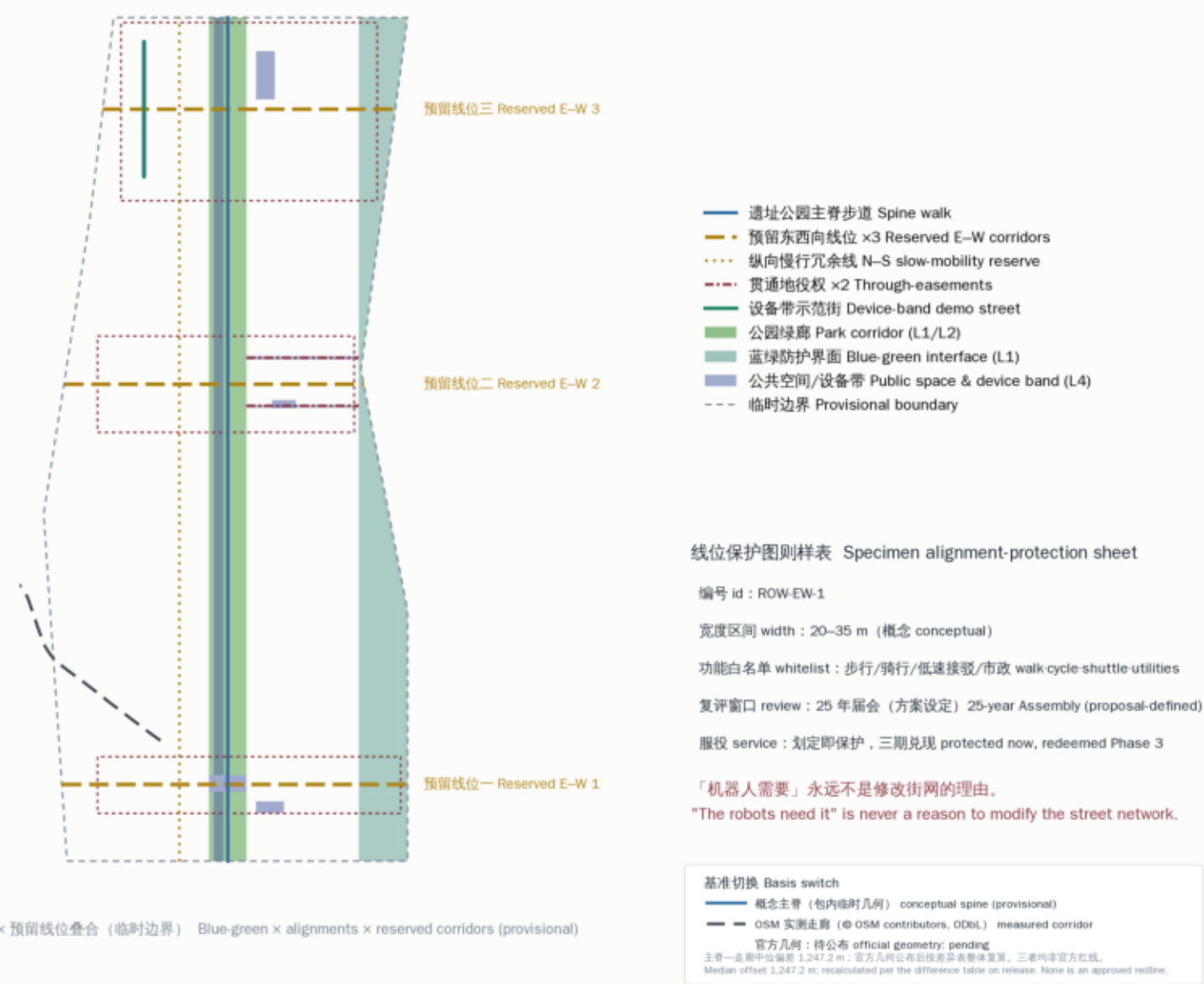
由本包 site_boundary / green_space / public_space 几何在 EPSG:4548 下复算；临时边界口径，低置信度。
Recomputed from this package's geometry in EPSG:4548; provisional-boundary basis, low confidence.

第一指标 slow_layer_change_count = 0 (恒)
The first metric: slow_layer_change_count = 0, always.

The Alignment Atlas

线位图则

交通慢行与蓝绿复合系统 Mobility & Blue-Green Composite



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► 3 reserved E–W corridors: Dazhongsi / Origin / Zhongzhiyuan

► 1 N–S slow-mobility redundancy line, parallel to the spine

► Per sheet: id / width range / whitelist / service dates / review window

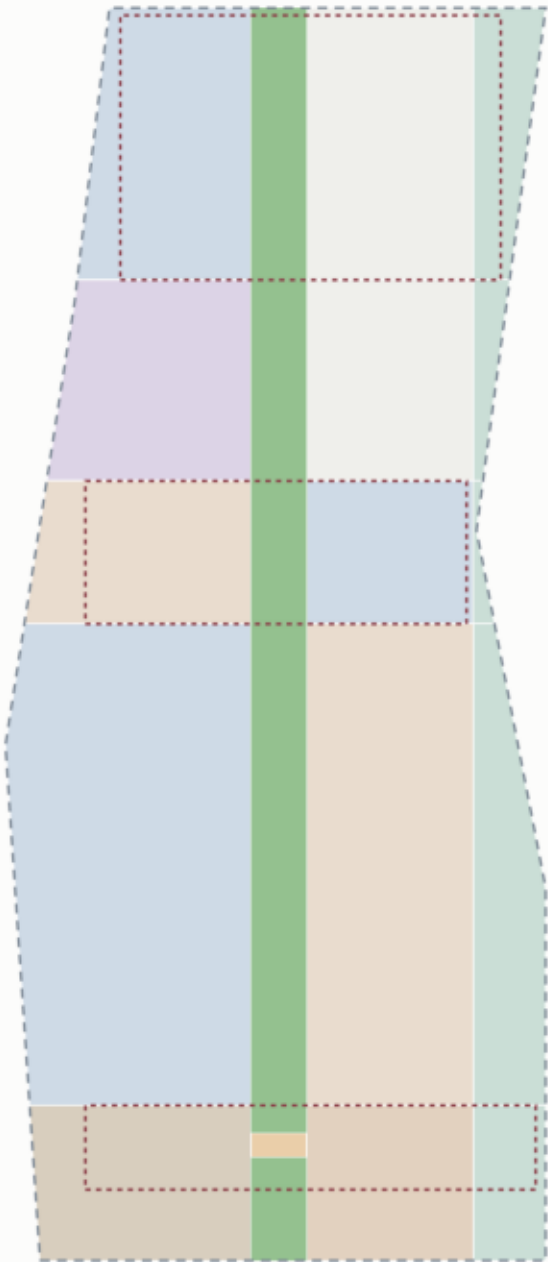
► Review window: the 25-year Assembly only (proposal-defined mechanism)

► All conceptual corridors, not engineering alignments (A-ROW-001)

Layered Control Guidance

分
层
管
控
引
导

分 层 管 控 引 导 图 Layered Control Guidance



用地图例 Land-use legend

- 0802 科研 Research
- 0701 居住 Residential
- 0804 教育 Education
- 0902 商务 Business
- 0901 商业 Commercial
- 16 留白 Reserve
- 1401 公园绿地 Park green
- 1402 防护绿地 Protective green
- 1403 广场 Plaza

分 层 管 控 引 导 一 览 Layered control guidance

每条指标 = 管控内容 + 锁定层位 + 复评周期
Each indicator carries a lock layer and a review period

- 蓝绿保护线 Blue-green line
锁定 lock: L1 复评 review: 不可议 Not amendable
- 线位与通行权 Alignments & ROW
锁定 lock: L2 复评 review: 25 年届会 25-y Assembly
- 贴线率/街墙 Build-to & street wall
锁定 lock: L3 复评 review: 50 年 50 y
- 高度分区骨架 Height zoning skeleton
锁定 lock: L3 复评 review: 50 年 50 y
- 街坊颗粒上限 Block-grain ceiling
锁定 lock: L3 复评 review: 50 年(听证) 50 y (hearing)
- 屋顶/第五立面 Roof & fifth facade
锁定 lock: L4 复评 review: 30 年 30 y
- 弹性断面/设备带 Sections & device band
锁定 lock: L4 复评 review: 15–30 年 y
- 广告与临时装置 Ads & installations
锁定 lock: L5 复评 review: 年度 Annual

容积率/高度/密度/绿地率/退线 : 待正式控规条件公布
FAR/height/density/green ratio/setback: pending official controls

Every control indicator
= content + lock layer
+ review period.

FAR/height etc. await
official conditions;
no approved values.

用地镶嵌概念模型（临时边界） Conceptual land-use mosaic (provisional)

Zhongzhiyuan: Fastest Layer, Most Conservative Chassis

众智园：最快的层，最保守的底盘

- The L5 test zone · 192.1 ha (provisional)
- Speed segregation: an L2-grade green corridor separates test from public
- However wild the tests, the chassis does not move
- Test transcripts price public-street lease deposits (S01 channel)
- Chassis redundancy: conduit headroom + elastic sections + device band

三区五速角色 Three Key Areas × Five-Speed Roles



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Zhongzhiyuan: The Device-Band Demonstration Street

众智园：设备带示范街

- The belt's first full device-band street, ~1.55 km (R-04)
- A 0.8–1.5 m reversible curbside strip: the only legal L5 landing
- Ledger plates: layer / term / deposit / restoration / parity path
- Unleased for two cycles → restored (the street obeys default exit too)
- Annual student co-creation leases (persona P5's interface)

Living-street elastic section (conceptual)



Origin Community: Winding the Block Clock

原点社区：拨动块钟

► The L3 pilot · 104.3 ha (provisional)

► Grain subdivision of an unnamed campus cluster (illustrative)

► Never compels stock; rights-holders decline → next candidate (R-06)

► Two through-easements (Civil Code arts. 372–385)

► Youth housing on new blocks: the L3 dividend to the youngest

Before: one 30-ha low-permeability compound

After: 4–6 small blocks + two 24 h passages
(illustrative; no real compound)

Origin Community: The Lease Service Station

原点社区：租约服务站

- ▶ Ledger plates, deposits, filings in one stop: the citizen-visible institution
- ▶ Deposit custody: licensed institution + regulatory compliance (pending)
- ▶ Five lease elements: layer/term/deposit/restoration/parity path
- ▶ Default exit: no renewal = automatic removal
- ▶ New professions: lease examiners, Assembly secretariat (slow-layer services)

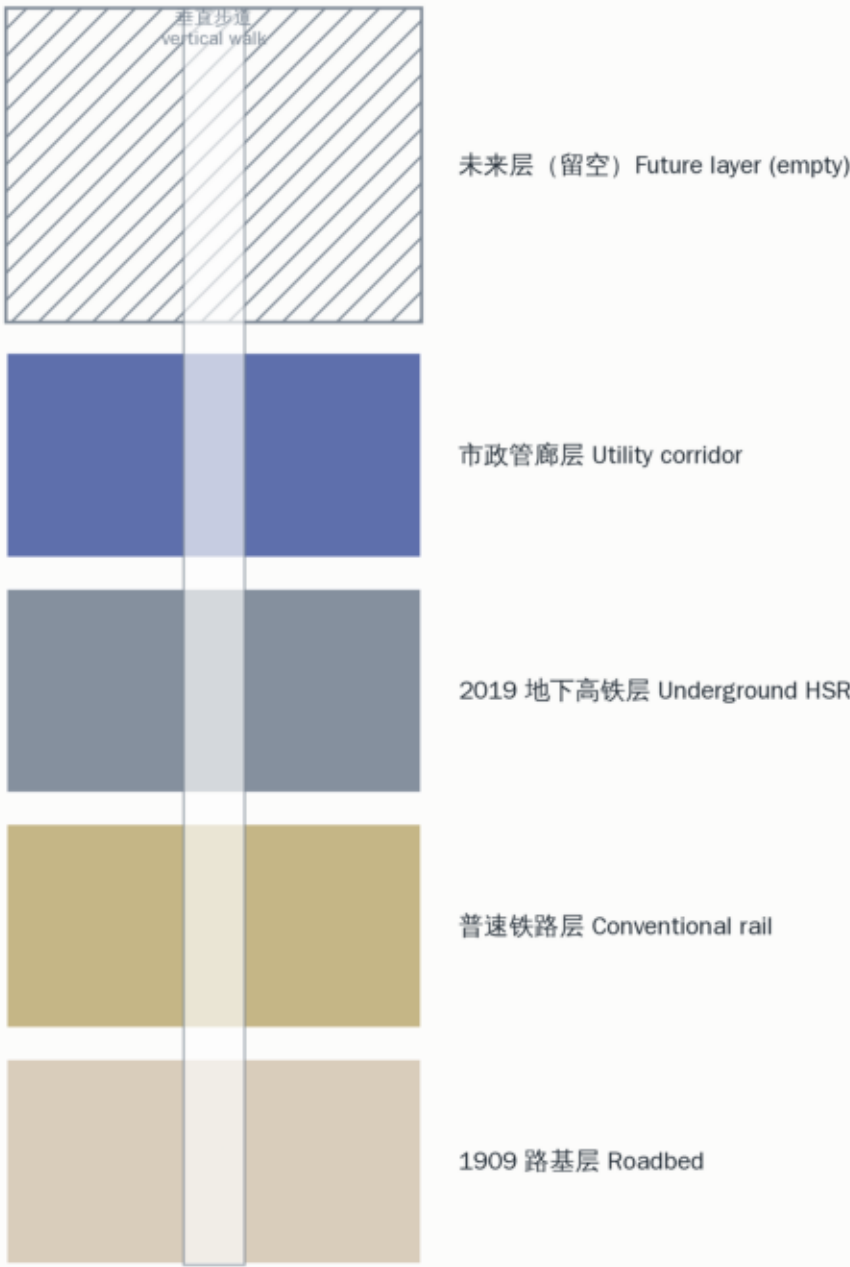
Dazhongsi: The Section Tower

大钟寺：断面塔

- ▶ An enterable vertical museum: one alignment, six technology strata
- ▶ The top "future layer" left deliberately empty
- ▶ Curatorial concept: no engineering feasibility conclusions
- ▶ Supports kept + curatorial infill (the King's Cross reading)

断面塔（概念剖切） Section Tower (conceptual cutaway)

策展性构筑物：不含桥隧/地下工程可行性结论
Curatorial structure: no engineering feasibility conclusions



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Dazhongsi: Five Clocks Plaza

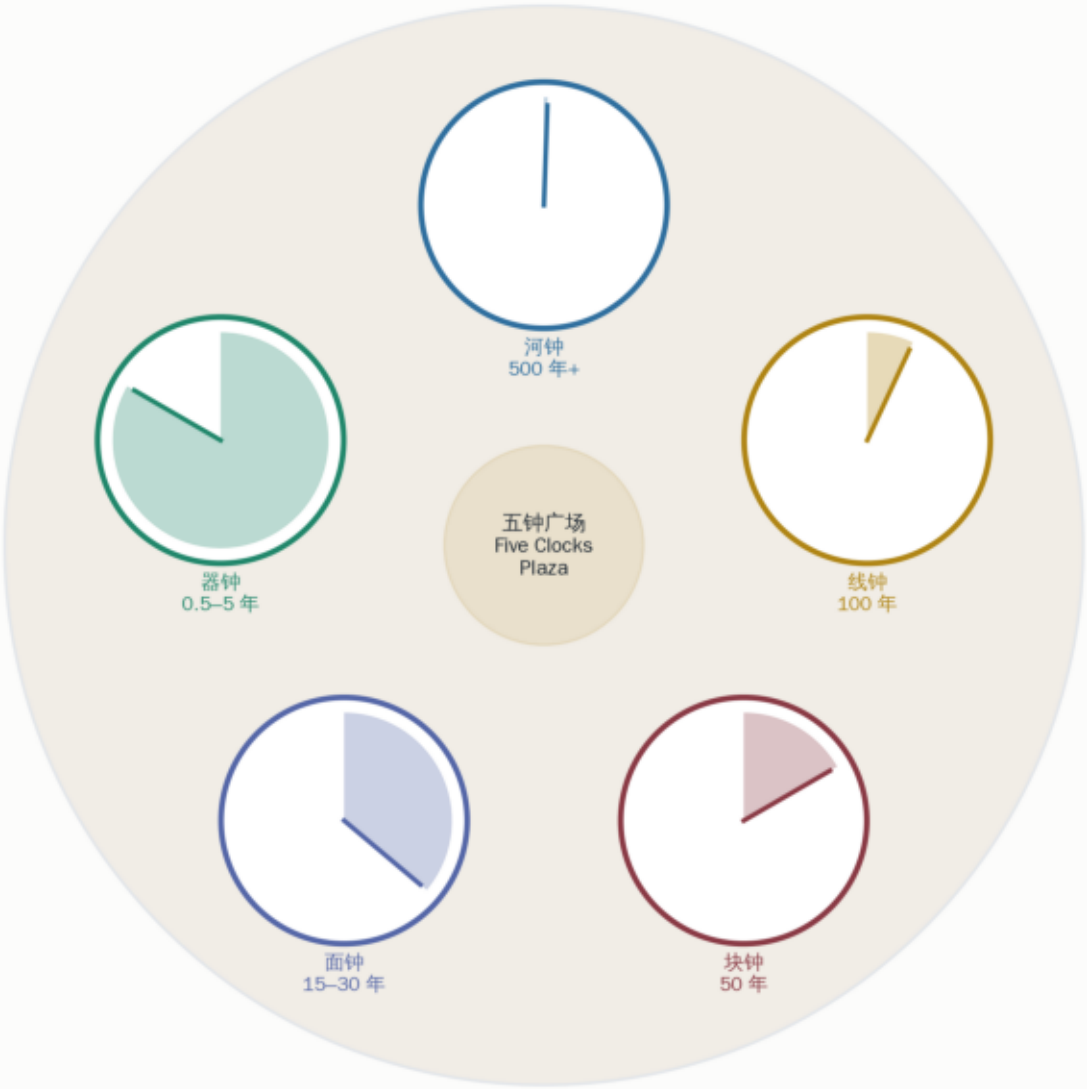
大钟寺：五钟广场

- Five city clocks, each at its layer's speed
- The Device dial spins; the River dial barely moves
- Public art + science communication, never entertainment
- Timeline paving terminus: a year per 50 m; future years empty
- Covenant stele: honor rings, one per 25 years

五钟广场平面（概念） Five Clocks Plaza plan (conceptual)

五口城市之钟各按层速运转：器钟快转，河钟近乎不动。公共艺术+科普定位。

Five city clocks, each at its layer's speed: the Device dial spins; the River dial barely moves. Public art + science communication.



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Scenario Cards, Selected (all 12 in the report)

场景卡精选 (12 卡全集见正文)

S01 Low-speed delivery robots

S01 低速配送机器人

L4 + L2 time-share · 2 y · human fallback in snow

L4+L2 分时 · 2 年 · 雪雨天降级人工

S08 Night cleaning (a pure time lease)

S08 夜间清扫 (纯时间租约)

Leases the spine's night hours — time is a layer

租的是主脊的夜间时段——时间也是层

S09 Waste-heat interface (classification demo)

S09 算力余热接口 (判层示范)

Looks AI, files as L4 → a 10-year term

貌似 AI 实属 L4 → 10 年租期

S12 Annual Chassis Day

S12 年度底盘日

One hour of L5 silence audits the parity paths

全带 L5 静默 1 小时，审计等价路径

All deposits illustrative; a non-AI parity path is statutory in every lease.

Projects and Phasing: R-01 ∙∙ R-10

项目与分期：R-01 ∙∙ R-10

一期 0–1 年 Phase 1

零施工 zero construction

R-01 plans · R-02 census · R-05 lease system · R-07 curation · R-10 baseline

R-01 图则 · R-02 颗粒普查 · R-05 租约系统 · R-07 策划 · R-10 基线档案

二期 1–5 年 Phase 2

可逆 reversible

R-04 demo street · R-08 paving 1 km · R-09 conduit assessment

R-04 示范街 · R-08 年表带一期 · R-09 管廊评估

三期 5–25 年 Phase 3

择机 as opportunity allows

R-03 corridor redemption · R-06 grain pilot

R-03 线位/地役权兑现 · R-06 颗粒试点

四期 25 年+ Phase 4

制度 institutional

Assembly cycle: 2051 · 2076 · 2101 · 2126

复线大会周期运转：2051 · 2076 · 2101 · 2126

Every project carries an exit; chains ≤ 2 ; R-10 seals this open call as the 2051 control group.

Metrics and Compliance

指标与合规

- First metric: slow_layer_change_count = 0, always
- site_area 11,412,825 m² · green 21.6% · public 4.0% (provisional)
- Computed: grain baseline (R-02, 2026-08-21 OSM); pending: conduit headroom (R-09), operations
- Coverage: announcement 1.3/1.4/1.5 + agent.1–agent.6
- Provisional boundary flagged in five places
- All spatial conclusions are conceptual recommendations

指标—证据链 Metrics—Evidence Chain

23 核心指标 · 已知/待复算 Core indicators · known vs pending		
slow_layer_change_count	0 count	●
reserved_row_count	4 count	●
reserved_row_length_m	13150.9 m	●
easement_passage_count	2 count	●
easement_coverage_ratio	待复算 pending	●
superblock_area_share	0.8478 ratio	●
median_block_area_ha	0.662 ha	●
superblock_count	66 count	●
ew_crossing_count	8 count	●
spine_to_corridor_offset_m	1247.2 m	●
device_band_coverage_km	1.565 km	●
corridor_conduit_headroom_years	待复算 pending	●
section_redundancy_ratio	待复算 pending	●
lease_active_count	待复算 pending	●
lease_return_rate	待复算 pending	●
lease_default_exit_rate	待复算 pending	●
deposit_pool_balance	待复算 pending	●
non_at_parity_coverage	1 ratio	●
annual_silent_drill_completion	待复算 pending	●
centennial_review_interval_years	25 years	●
timeline_paving_length_m	1000 m	●
speed_diversity_index	待复算 pending	●
component_library_items	6 count	●
● 实心 = 已知 (几何/设计声明/OSM 普查复算) ○ 灰 = 待正式数据/运营数据 filled = known (geometry / design declaration / OSM census) · grey = pending official/operational data		

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